

CODEALPHA INTERNSHIP

TASK 3

E-Commerce Sales Data Visualization

Data Visualization using Python, Pandas and Matplotlib

1. Introduction

This project presents a data visualization analysis of an e-commerce sales dataset. The objective is to transform sales data into clear visual representations that make revenue trends, category performance, city-level performance, product performance, quantity-based revenue and quarterly patterns easier to understand.

2. Objectives

- Create meaningful visualizations from the e-commerce sales dataset.
- Identify revenue trends across months and quarters.
- Compare revenue performance across product categories and cities.
- Identify the top products by revenue.
- Understand how revenue varies with order quantity.
- Present the results through a dashboard-style visualization.

3. Dataset Overview

The dataset contains 1,500 e-commerce sales records with 12 variables: Order_ID, Order_Date, City, Category, Product, Quantity, Unit_Price, Discount, Revenue, Month, Month_Number and Quarter.

4. Tools and Technologies

- Python
- Pandas – data loading, grouping and aggregation
- Matplotlib – charts and visualizations
- Jupyter Notebook – development and presentation environment
- Excel-compatible spreadsheet – source dataset

5. Visualization Methodology

The dataset was loaded into a Pandas DataFrame. Revenue was aggregated using groupby operations according to the required dimensions. Matplotlib was used to create line and bar charts. The visualizations were designed to compare revenue and identify patterns in the sales data.

6. Dashboard-Style Visualization

The dashboard combines monthly revenue trend, revenue by category, revenue by city and revenue by quarter in a single visual view.

```
plt.tight_layout()
plt.show()
```

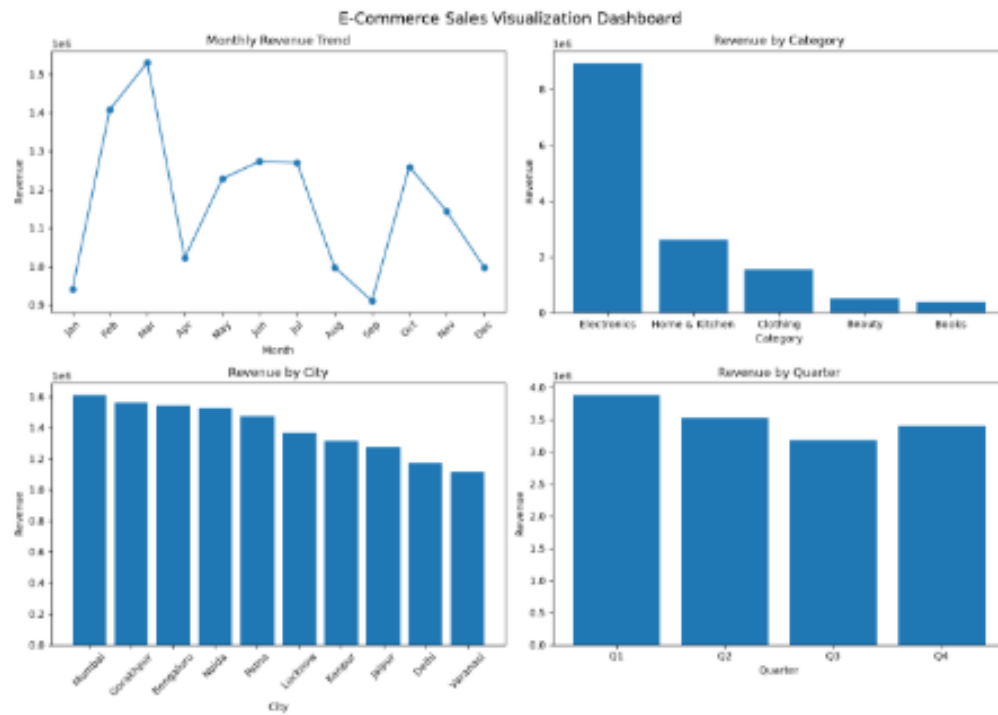


Figure 1: E-Commerce Sales Visualization Dashboard

7. Monthly Revenue Trend

A line chart was used to visualize revenue across January to December. March recorded the highest monthly revenue at ₹15,30,403.15, while September recorded the lowest monthly revenue at ₹9,10,548.90.



Figure: Original visualization from the supplied Task 3 dashboard.

8. Revenue by Category

A bar chart was used to compare total revenue generated by product category. Electronics generated the highest category revenue, with ₹89,37,601.40.

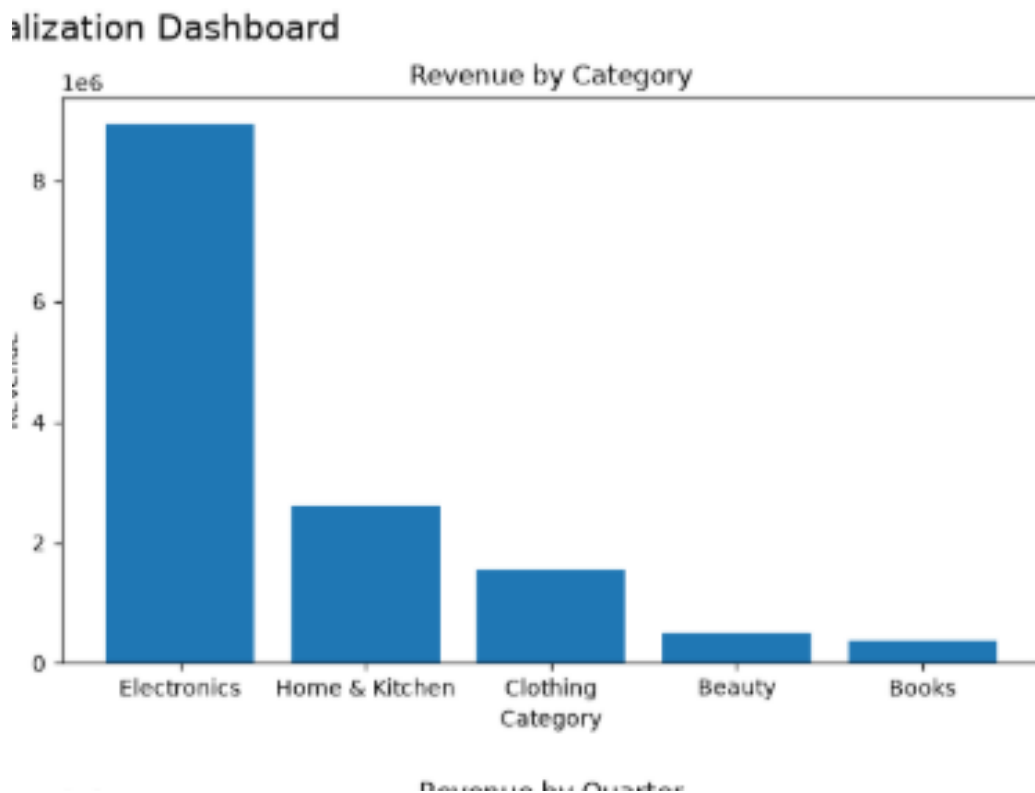


Figure: Original visualization from the supplied Task 3 dashboard.

9. Revenue by City

A city-wise bar chart was created to compare total revenue across cities. Mumbai recorded the highest city-level revenue at ₹16,08,805.10.

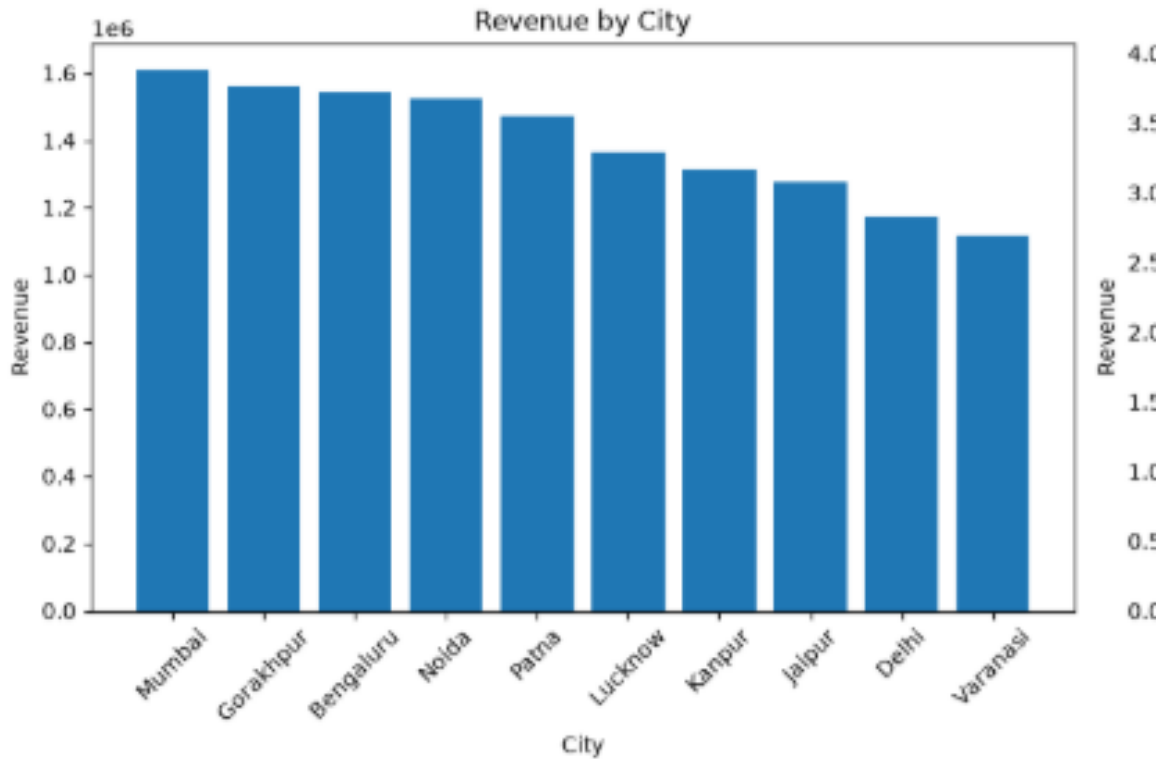


Figure: Original visualization from the supplied Task 3 dashboard.

10. Top 10 Products by Revenue

A horizontal bar chart was created to display the ten products with the highest total revenue. Power Bank was the highest-revenue product in the analysis, with ₹21,21,624.25.

11. Revenue by Quantity

A bar chart was used to compare total revenue for orders with quantities 1, 2, 3 and 4. This helps examine how revenue is distributed across quantity levels.

12. Revenue by Quarter

A quarterly bar chart was created to compare revenue across Q1, Q2, Q3 and Q4, providing a compact view of quarterly sales performance.

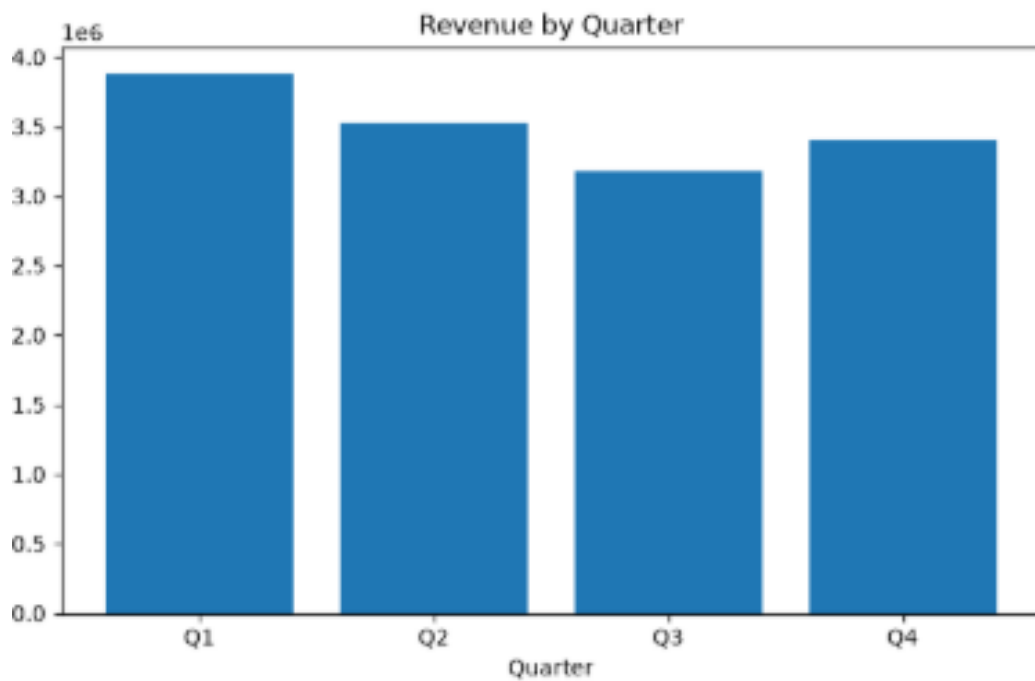


Figure: Original visualization from the supplied Task 3 dashboard.

13. Key Insights

- Monthly revenue varies across the year.
- March recorded the highest monthly revenue and September the lowest.
- Electronics was the highest-revenue category.
- Mumbai generated the highest city-level revenue.
- Power Bank generated the highest product-level revenue.
- The quarterly chart provides a clear Q1–Q4 comparison.
- The dashboard combines multiple revenue perspectives in one visual view.

14. Conclusion

The project demonstrates how e-commerce sales data can be converted into meaningful visual insights using Python, Pandas and Matplotlib. The combination of trend, comparison and dashboard-style charts makes the dataset easier to interpret and communicate.

15. Project Files

- CodeAlpha_ECommerce_Visualization.ipynb – Jupyter Notebook containing the analysis and visualization code.
- CodeAlpha_ECommerce_Sales_Dataset.xlsx – source dataset.
- CodeAlpha_Task_3_Data_Visualization_Report.docx – project report.

Complete E-Commerce Sales Visualization Dashboard

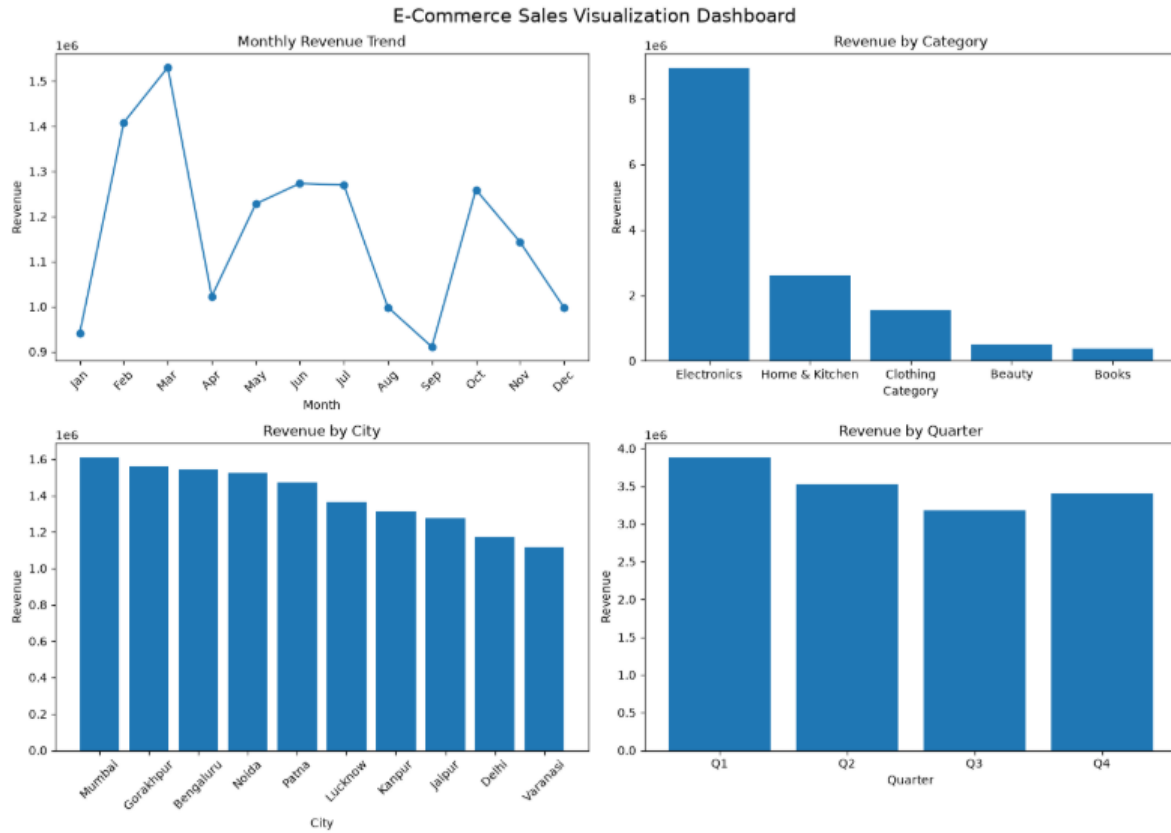


Figure: Complete original dashboard supplied for CodeAlpha Task 3.